Model-Agnostic Interpretation of Machine Learning via Average Marginal Effects

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Abstract

We propose a framework for interpreting any supervised machine learning model through average marginal effects, the standard inferential tool in the econometrics of nonlinear models. The estimand is the window average marginal effect, the average centred finite difference of the regression function at a fixed, interpretable step. It is well defined for nowhere-differentiable learners and recovers the classical average derivative as the window shrinks. Because the functional is linear in the regression function, it admits an exact Riesz representation obtained by a change of variables, with no boundary terms or density derivatives, and the resulting orthogonal score has estimation bias exactly equal to a product of two squared-error norms. A cross-fitted debiased estimator is therefore root-n consistent and asymptotically normal whenever the learner and a Riesz-regression estimate of the representer converge faster than the fourth root of the sample size; it attains the semiparametric efficiency bound and returns a coefficient table analogous to parametric regression output without resampling. The score is also doubly robust. A pairing identity characterizes what the marginal effect of a tree-based model measures, and gradient consistency results for neural networks and Mondrian forests delineate the narrower regime where an uncorrected plug-in is already consistent.

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1 Introduction

The rise of flexible machine learning methods has created a persistent tension in empirical research. Models such as random forests (Breiman, 2001b), gradient boosted trees (Chen and Guestrin, 2016), and deep neural networks (Hastie et al., 2009) routinely achieve predictive accuracy that parametric models cannot match. Yet the standard tools of statistical inference — coefficient estimates, standard errors, hypothesis tests — apply naturally only to parametric models whose functional form is specified in advance. Breiman (2001a) framed this as a conflict between two cultures of statistical modelling: one oriented towards interpretable data-generating models, the other towards accurate algorithmic prediction. The conventional resolution has been to treat accuracy and interpretability as opposing ends of a spectrum, accepting reduced interpretability as the price of improved prediction. This paper argues that the tradeoff is not fundamental. It is an artefact of requiring parametric assumptions to make interpretation rigorous, and it disappears once that requirement is dropped.

The machine learning literature has produced a rich set of post-hoc interpretability methods that attempt to explain black-box model predictions without parametric assumptions. Partial dependence plots (Friedman, 2001) visualize the marginal relationship between a feature and the predicted outcome by averaging over the empirical distribution of other features. LIME (Ribeiro et al., 2016) fits a local linear approximation to the prediction function in the neighbourhood of individual observations. SHAP (Lundberg and Lee, 2017) assigns each feature a contribution to each prediction based on Shapley values from cooperative game theory, providing a unified framework that nests several earlier methods. These tools are widely used in practice and are surveyed comprehensively in Molnar (2025) and Doshi-Velez and Kim (2017). They share a fundamental limitation, however: they produce point estimates with no associated sampling distribution. A SHAP value, a PDP slope and a LIME coefficient are descriptive quantities, carrying no standard error, no confidence interval and no basis for a hypothesis test, so a researcher cannot say whether an estimated effect is distinguishable from zero or whether it differs across model specifications.

Two prior contributions are closest in spirit to the present paper. Scholbeck et al. (2024) propose forward marginal effects for nonlinear prediction functions as a model-agnostic interpretability method, establishing a general framework for computing feature effects for any supervised learner. Their framework is descriptive rather than inferential — no standard errors, no confidence intervals, no tests — and it explicitly argues against summarizing feature effects in a single metric such as the average marginal effect. The present paper takes the opposite position: the average marginal effect is the right summary quantity, it carries a population-level interpretation inherited from the econometric literature, and semiparametrically efficient inference for it is achievable with a debiased, cross-fitted estimator and an explicit influence function. Löwe et al. (2024) provide the `fmeffects` R package implementing that descriptive framework, which demonstrates the practical demand for marginal-effect-style interpretation without supplying the population estimand or the efficiency theory developed here.

The econometric literature has solved exactly this inferential problem for parametric models through average marginal effects (Greene, 2012; Wooldridge, 2010). For a generalized linear model with inverse link function μ and linear predictor $\eta = \mathbf{x}^\top \boldsymbol{\beta}$, the average marginal effect of variable j is $\text{AME}_j = \beta_j \cdot \mathbb{E}[\mu'(\eta)]$ — the regression coefficient scaled by the average derivative of the inverse link, averaged over the empirical distribution of the linear predictor. reducing to the OLS coefficient under the identity link. The result is measured in the natural units of the outcome and interpretable as the average change in the predicted outcome per one-unit change in x_j, with standard errors from the delta method or the bootstrap. This is the output of `margins` in R and Stata (Leeper, 2021), and the standard form in which marginal effects are reported across the applied social sciences. The limitation is that it has been restricted to parametric models: to compute $\partial f / \partial x_j$ analytically, one must know the functional form of f.

A separate literature, largely disjoint from both the interpretability and the applied marginal effects traditions, has studied the population version of this quantity directly. The average derivative $\mathbb{E}[\partial_j f(X)]$ was introduced as an estimand in its own right by Härdle and Stoker (1989) and Powell et al. (1989), its semiparametric efficiency bounds

were derived by Newey and Stoker (1993), and the modern debiased machine learning programme (Chernozhukov et al., 2018, 2022) shows how to estimate linear functionals of a regression function at the parametric rate using flexible learners as nuisance estimators. The present paper is deliberately positioned at the junction of these literatures, and its delta relative to each is stated in one sentence: the debiasing literature tells one how to estimate $\mathbb{E}[\partial_j f(X)]$ efficiently when the derivative exists and the learner is treated as a black-box nuisance; this paper establishes what the estimand *is* when the learner is not differentiable — the window effect, with an exact finite-difference Riesz representation requiring no boundary conditions and no density derivatives — and shows that a single orthogonal score then covers smooth and piecewise constant learners, continuous and discrete regressors, within one construction. Conversely, relative to the interpretability literature, the contribution is to replace descriptive scores with an estimand–estimator–efficiency-bound triple of the classical kind.

This paper removes that restriction. We propose a framework for computing average marginal effects for any supervised machine learning model, requiring only that the prediction function $\hat{f}$ can be evaluated at a point. The estimand is the *window average marginal effect*: the expectation of a centred finite difference of the regression function at a fixed step h scaled to the spread of each feature, trimmed away from the boundary of the support. Treating the finite difference as the estimand rather than as a numerical approximation to a derivative is what makes genuine model-agnosticism possible — the window effect is well defined for tree-based models, for which classical derivatives do not exist anywhere — while the classical average derivative is recovered as the $h \to 0$ limit whenever the regression function has one weak derivative, so nothing is given up relative to the econometric tradition. Estimation combines the plug-in average of finite differences with an orthogonal correction term built from an explicit Riesz representer, computed by cross-fitting; standard errors come from the estimated influence function in a single pass, with no resampling required, and a multiplier bootstrap is available at no additional model fits when a resampling distribution is wanted. The result is a coefficient table — estimate, standard error, and significance level — directly analogous to the out-

put of a parametric regression but with no restrictions on the functional form of $\hat{f}$. The framework nests the classical GLM result as a special case: when $\hat{f}$ is a logistic regression, the estimator recovers the classical AME automatically. When $\hat{f}$ is XGBoost or a random forest or a neural network, the same formula applies and produces output on the same scale and with the same interpretation, and for binary regressors the construction specializes to augmented inverse probability weighting with the regressor in the role of the treatment.

The theoretical contribution proceeds in three parts, answering in turn the questions of inference, optimality and interpretation. First, because the window functional is linear in the regression function, a change of variables yields an exact Riesz representation with an explicitly computable representer — no integration by parts, no boundary terms, and no derivatives of the covariate density — and the bias of the resulting orthogonal moment is exactly a product of two L^2 estimation errors. A cross-fitted estimator built on that score is root-n consistent and asymptotically normal whenever the learner and a Riesz-regression estimate of the representer each converge faster than $n^{-1/4}$ in L^2, in place of the parametric-rate function-space convergence that a delta-method treatment would demand and that no growing-complexity learner satisfies. Second, the score is the efficient influence function, so the estimator attains the semiparametric efficiency bound for the window effect; it is moreover doubly robust, remaining valid with no consistency requirement on the learner at all whenever the representer is known or parametrically estimable, which covers the fixed-complexity defaults under which off-the-shelf learners converge to pseudo-true limits rather than to the truth. Third, a pairing identity characterizes what the marginal effect of a non-differentiable model measures: for any estimator of bounded variation — every tree-based model — the window effect equals the jump measure of the fitted surface integrated against a locally averaged covariate density. Gradient consistency is established for neural networks in Sobolev space and for gradient boosted models and Mondrian forests (Lakshminarayanan et al., 2014; Mourtada et al., 2017) in the space of functions of bounded variation; for Breiman forests (Breiman, 2001b; Scornet et al., 2015) it is conjectured rather than proved, with the obstruction identified. Because the

inference theory requires gradient consistency of no class, valid inference for a Breiman forest needs only an L^2 rate, while the gradient question bears on interpretation rather than on coverage and remains open.

We validate the framework on three datasets spanning binary classification in social science, binary classification in finance, and continuous regression: the UCI Adult Income dataset (Kohavi, 1996; Dua and Graff, 2019), the UCI Credit Card Default dataset (Yeh, 2009; Dua and Graff, 2019), and the Ames Housing dataset (De Cock, 2011). In each we compare average marginal effects from logistic or linear regression, random forests, XGBoost (Chen and Guestrin, 2016) and neural networks against SHAP values (Lundberg and Lee, 2017) and partial dependence plot slopes (Friedman, 2001). The debiased effects agree closely across the four model classes, which is what one expects once every learner is used to estimate the same window effect rather than to report its own slope; SHAP values answer a different question and are correspondingly on a different scale and sometimes of opposite sign. The credit default application is the sharpest case and is worth stating at the outset. Under the plug-in the four learners put the effect of the payment ratio anywhere from -0.440 to 0.0009 — a set of estimates that cannot all be describing the same world, and among which a reader had no principled way to adjudicate. Under the correction all four are negative and substantial. The disagreement was not four models detecting different structure; it was the regularization bias $\mathbb{E}[\alpha_h(\hat{f} - f)]$, which differs across learners because their regularization differs, and which the orthogonal score removes.

An additional finding concerns Breiman random forests, the learner the framework is most often asked to carry and the one for which the interpretation theory is least complete. Under the uncorrected plug-in they behave worst of the four classes: the simulation study shows a bias that grows with the sample size rather than vanishing, so intervals shrink at the usual rate around a centre that is not moving and coverage collapses to 0.130. The orthogonal correction removes that term and returns coverage to a mean of 0.948 against the nominal 0.95 across all one hundred and twenty forest cells of the design. The reason this is more than a bias correction matters for which learners

the framework can be used with. A depth-constrained forest whose plug-in error grows with n is not converging to f, and Proposition 2 says that with the representer known this costs nothing: the estimator is root-n normal about $\theta_h(f)$ even when $\hat{f}$ converges to some other function entirely. Because the inference theory consumes gradient consistency for no learner class, the Breiman forest is a natural choice for applied work rather than a concession.

The remainder of the paper is organized as follows. Section 2 defines the estimand and the estimator and establishes their relationship to classical average marginal effects for the generalized linear model family. Section 3 develops the asymptotic theory. Section 4 presents the credit default application and Section 5 the simulation evidence; the other two applications, the detailed simulation results and the proofs of every result stated in Section 3 are in the supplement. Section 6 concludes.

2 Method

We propose a framework for interpreting any supervised machine learning model through the lens of average marginal effects. The framework is agnostic to the choice of learner — the same estimation procedure applies whether $\hat{f}$ is a neural network, a gradient boosted model, a random forest, or any other supervised learner. Crucially, the framework produces standard errors from the influence function of a Neyman-orthogonal score, evaluated by cross-fitting, enabling hypothesis tests and confidence intervals that existing model-agnostic interpretability methods do not support. A refitting bootstrap is retained, but as a diagnostic rather than as the basis for inference: Section 3.4 shows that it is blind to the learner's regularization bias, which is precisely the term the orthogonal correction removes. SHAP values and permutation-based feature importance measures provide point estimates of variable importance but no basis for inference. The average marginal effect estimator proposed here yields a coefficient table — estimate, standard error, and significance — directly analogous to the output of a parametric regression, with no restrictions on the functional form of $\hat{f}$.

2.1 Estimand

Let $\mathbf{x}_i = (x_{i1}, \ldots, x_{ip})^\top \in \mathbb{R}^p$ denote the feature vector for observation i, and let $y_i \in \mathbb{R}$ denote the outcome. We observe n independent draws $\{(\mathbf{x}_i, y_i)\}_{i=1}^n$ from some population distribution P. Let $f : \mathbb{R}^p \to \mathbb{R}$ denote the conditional expectation function $f(\mathbf{x}) = E[Y \mid \mathbf{x}]$. For binary outcomes, $f(\mathbf{x}) = P(Y = 1 \mid \mathbf{x})$ is the conditional probability of the positive class; for continuous outcomes, $f(\mathbf{x}) = E[Y \mid \mathbf{x}]$ directly. In both cases f is the same object — the conditional expectation of Y given $\mathbf{x}$ — and the method requires no modification across outcome types.

The marginal effect of variable j at observation i is the partial derivative of f evaluated at $\mathbf{x}_i$:

$$m_{ij} = \left.\frac{\partial f}{\partial x_j}\right|_{\mathbf{x}_i} \tag{2.1}$$

The population average marginal effect of variable j is:

$$\mathrm{AME}_j = \mathbb{E}\left[\frac{\partial f}{\partial x_j}(\mathbf{x})\right] \tag{2.2}$$

where the expectation is taken over the marginal distribution of $\mathbf{x}$ under P. This is the classical target. The operational estimand of the paper is its finite-window counterpart, the *window average marginal effect* $\theta_{j,h} = \mathbb{E}[w(\mathbf{x})\,\{f(\mathbf{x} + he_j) - f(\mathbf{x} - he_j)\}/2h]$, where $h > 0$ is a fixed step scaled to the spread of x_j and w is a trimming weight that excludes points within h of the boundary of the support. The window effect is well defined for any bounded prediction function — no derivative need exist anywhere, which is what a genuinely model-agnostic estimand requires — and it converges to AME_j as $h \to 0$ whenever f has one weak derivative. Both facts, and the conditions under which the estimators below converge, are established in Section 3.

2.2 Estimation

Given an estimator $\hat{f}$ of f, the sample analog is:

$$\widehat{\mathrm{AME}}_j = \frac{1}{n}\sum_{i=1}^{n} \frac{\partial \hat{f}}{\partial x_j}\bigg|_{\mathbf{x}_i} \tag{2.3}$$

For most supervised learners the partial derivative $\partial\hat{f}/\partial x_j$ is not available in closed form, and for tree-based models it does not exist everywhere in the classical sense. We therefore estimate it via a centred finite difference:

$$\frac{\partial \hat{f}}{\partial x_j}\bigg|_{\mathbf{x}_i} \approx \frac{\hat{f}\left(\mathbf{x}_i^{(j,\, x_{ij}+h_j)}\right) - \hat{f}\left(\mathbf{x}_i^{(j,\, x_{ij}-h_j)}\right)}{2h_j} \tag{2.4}$$

where $\mathbf{x}_i^{(j,v)}$ denotes the vector $\mathbf{x}_i$ with the j-th component replaced by v. We write $\Delta_{j,h}g(\mathbf{x})$ for the right-hand side of (2.4) applied to a generic function g, so that the sample analog above is $n^{-1}\sum_i \Delta_{j,h}\hat{f}(\mathbf{x}_i)$; the same notation is used throughout Section 3. The centred finite difference is preferred over a one-sided difference because its approximation error relative to the pointwise derivative is $O(h_j^2)$ rather than $O(h_j)$, which becomes material when h_j is not negligibly small. Relative to the window estimand $\theta_{j,h}$ the centred difference involves no approximation at all: it is the defining operation of the estimand, computed exactly, and h_j is a reported feature of the target — the perturbation scale at which effects are measured — rather than a numerical tolerance.

The step size h_j is chosen adaptively as:

$$h_j = \max\bigl(10^{-4},\ 0.05 \cdot \hat{\sigma}_j\bigr) \tag{2.5}$$

where $\hat{\sigma}_j$ is the sample standard deviation of the j-th feature. Scaling h_j to the spread of x_j ensures the finite difference is neither so small as to introduce numerical instability nor so large as to obscure local behaviour. The floor of 10^{-4} guards against degenerate cases where $\hat{\sigma}_j \approx 0$. Observations within h_j of the boundary of the observed support receive trimming weight zero, so that $\hat{f}$ is never evaluated at points where the regression function is unidentified; the trimmed shell carries probability mass of order h_j, and with

the default step this exclusion is small. Untrimmed evaluation — the default in earlier implementations — silently extrapolates the fitted model outside the support, and the trimming weight should be understood as the repair of that extrapolation rather than as a restriction of the method.

For binary or categorical variables the partial derivative does not exist and the finite difference is replaced by a contrast. The marginal effect for observation i is:

$$m_{ij} = \hat{f}\left(\mathbf{x}_i^{(j,1)}\right) - \hat{f}\left(\mathbf{x}_i^{(j,0)}\right) \tag{2.6}$$

which measures the change in predicted outcome when x_j is switched from 0 to 1 holding all other variables at their observed values. The sample average marginal effect is then $\widehat{\text{AME}}_j = n^{-1} \sum_{i=1}^{n} m_{ij}$, with interpretation identical to the continuous case: a change in x_j from 0 to 1 is associated with a $\widehat{\text{AME}}_j$ change in the predicted outcome, on average across the sample.

For binary classification models, $\hat{f}$ must return the estimated conditional probability $\hat{P}(Y = 1 \mid \mathbf{x})$ rather than the predicted class label. Class labels are discrete and their finite differences are identically zero; the method requires a continuous output. This is the predicted probability of the positive class, available as `predict_proba(X)[:,1]` in scikit-learn and equivalent interfaces in other frameworks. The AME is then interpreted as the change in predicted probability associated with a one-unit change in x_j, on average across the sample.

2.3 Relationship to Classical Average Marginal Effects

Average marginal effects are a standard tool in the econometrics of nonlinear models (Greene, 2012; Wooldridge, 2010). For parametric models the estimator $\widehat{\text{AME}}_j$ defined above reduces exactly to the classical result. Consider a generalized linear model with inverse link function μ and linear predictor $\eta_i = \mathbf{x}_i^\top \boldsymbol{\beta}$, so that $f(\mathbf{x}_i) = \mu(\eta_i)$. The marginal

effect of variable j at observation i is:

$$\left.\frac{\partial f}{\partial x_j}\right|_{\mathbf{x}_i} = \beta_j \cdot \mu'(\eta_i) \tag{2.7}$$

and the population average marginal effect is $\text{AME}_j = \beta_j \cdot E[\mu'(\eta)]$. For the identity link $\mu'(\eta) = 1$ and $\text{AME}_j = \beta_j$ — the marginal effect is constant across observations and equals the OLS coefficient. For the logistic link $\mu'(\eta) = \Lambda(\eta)(1 - \Lambda(\eta))$ and the AME averages the logistic density over the sample (Greene, 2012). For the probit link $\mu'(\eta) = \phi(\eta)$, the standard normal density evaluated at the linear predictor (Wooldridge, 2010).

In each case $\widehat{\text{AME}}_j$ recovers the classical result automatically — the finite difference applied to the model output numerically approximates $\beta_j \cdot \mu'(\eta_i)$ without requiring knowledge of the link function. The framework therefore nests the entire GLM family as a special case and reduces to familiar output when the model is parametric. This backward compatibility is what the `marginaleffects` package in R and Python implements (Arel-Bundock et al., 2024); the present framework extends that implementation with formal asymptotic guarantees for machine learning models.

2.4 Inference

Inference is based on a debiased, cross-fitted estimator whose formal analysis is the subject of Section 3; here we describe the procedure. Partition the sample into K folds ($K = 5$ by default). For each fold k, using only the observations outside the fold: (1) fit the learner $\hat{f}^{(-k)}$ exactly as it would be fit in any application, tuning included if the learner is tuned, using only that fold's training portion; (2) fit an estimate $\hat{\alpha}^{(-k)}$ of the Riesz representer of the window effect — an explicit weight function characterized in Section 3 — by minimizing an empirical quadratic loss that requires only evaluating candidate functions at the shifted points $\mathbf{x}_i \pm h_j e_j$, with no density estimation of any kind. The estimate of $\theta_{j,h}$ averages, over the held-out observations of every fold, the finite difference

of the out-of-fold learner plus a correction term:

$$\hat{\theta}_{j,h} = \frac{1}{n} \sum_{k=1}^{K} \sum_{i \in I_k} \left[w(\mathbf{x}_i)\, \Delta_{j,h} \hat{f}^{(-k)}(\mathbf{x}_i) + \hat{\alpha}^{(-k)}(\mathbf{x}_i) \left(y_i - \hat{f}^{(-k)}(\mathbf{x}_i) \right) \right] \tag{2.8}$$

The first term is the plug-in average of Section 2.2 computed on held-out data; the second term removes, at first order, the bias that regularization of the learner transmits to the plug-in average — each held-out residual, weighted by the representer, testifies to how the learner is locally mis-calibrated where that mis-calibration matters for the marginal effect. For binary and categorical variables the same formula applies with the contrast in place of the finite difference and inverse propensity weights in the role of the representer.

Standard errors require no resampling: the summands above, centred at $\hat{\theta}_{j,h}$, are estimates of the influence function values, and the standard error is the sample standard deviation of those values divided by $\sqrt{n}$. The coefficient table — estimate, standard error, t-statistic, and p-value for each feature — is therefore produced by K model fits in total, in place of the hundreds of refits that a bootstrap requires; Section 3 establishes that the resulting confidence intervals are asymptotically valid and that the estimator attains the semiparametric efficiency bound. When a full resampling distribution is wanted — for simultaneous confidence bands across features, for example — a multiplier bootstrap that perturbs the estimated influence function values with i.i.d. mean-zero weights is available at no additional model fits.

The package additionally retains a nonparametric model-refitting bootstrap (Efron, 1979; Efron and Tibshirani, 1993) as a *diagnostic* for how far the fitted function moves when the sample is perturbed — variability that the asymptotic theory conditions away. The learner's settings are held fixed across replicates; if the learner is tuned, the tuning is part of what each replicate refits. For $b = 1, \ldots, B$: draw $\mathcal{D}^{(b)}$ by resampling $(\mathbf{x}_i, y_i)$ pairs jointly with replacement; refit $\hat{f}^{(b)}$ on $\mathcal{D}^{(b)}$ from a fresh initialization at the same settings; and recompute the plug-in average marginal effect. The dispersion of the replicates measures how sensitive the reported effects are to the joint variability of resampling and refitting, and large dispersion is a useful warning that the fitted model is unstable. It is

not, however, a substitute for the influence function standard errors above, for reasons Section 3 makes precise: the refitting bootstrap recenters at the fitted model and is therefore blind to the regularization bias that the correction term removes — a blindness whose consequences for coverage are documented for random forests in Section 5 — and, because each replicate is evaluated at the original sample, it omits the component of sampling variance that comes from averaging a heterogeneous effect over a finite sample of covariate values.

3 Asymptotic Theory

3.1 Overview

The theory must deliver three things: a population estimand that is well defined for any supervised learner, including learners that are nowhere differentiable; valid root-n inference on that estimand under conditions flexible learners can actually satisfy, in particular without requiring the learner itself to converge at the parametric rate, which fails whenever the complexity of the function class grows with the sample size; and an account of what the gradient of a fitted model measures. This section delivers them in turn.

The organizing observation is structural: the average marginal effect is a *linear* functional of the regression function, and linearity has three consequences that together carry the entire argument. First, the functional admits an exact Riesz representation — an explicit weight α_h against which averaging a function reproduces the average of its finite difference (Lemma 1) — obtained by a change of variables rather than integration by parts, so that no boundary terms arise and no derivative of the covariate density is ever needed. Second, the estimation bias of a debiased moment built from that representation is *exactly* a product of two estimation errors (Lemma 2), an identity rather than a product plus a linearization remainder as it would be for a nonlinear functional. Root-n inference therefore requires only that the product of the L^2 errors of two nuisance estimates be $o_p(n^{-1/2})$ — for instance, both converging faster than $n^{-1/4}$ — rather than root-n con-

vergence of the learner in any norm (Theorem 1). Third, the resulting estimator attains the semiparametric efficiency bound for the estimand (Theorem 2).

The estimand analysed here is exactly the quantity the estimator of Section 2 computes, with the step h treated as part of the estimand rather than as a numerical approximation to a derivative; what a learner's own gradient estimates is taken up in Section 3.5. The plug-in average is the first component of the orthogonal score, and the correction term $\alpha_h(x_i)\,(y_i - \hat{f}(x_i))$ removes, at first order, the bias that regularization of the learner transmits to it — the mechanism the simulations of Section 5 display directly.

3.2 Notation and Preliminaries

Let $\Omega \subset \mathbb{R}^d$ be a bounded Lipschitz domain, let $f : \Omega \to \mathbb{R}$ be the regression function, and let $\hat{f}_n$ be an estimator of f built from n observations $(\mathbf{x}_i, y_i)_{i=1}^n$, independent draws from P; with mild abuse of notation P also denotes the covariate marginal, with $\|g\|_{P,2}^2 = \mathbb{E}[g(X)^2]$. Boundedness and Lipschitz regularity of Ω are what make the compactness and embedding results of Section S2 available, and hold for any compact covariate space with reasonable boundary behaviour. We fix a coordinate $j \in \{1, \dots, d\}$ for the remainder of the section and suppress it where no confusion arises. The centred difference operator $\Delta_{j,h}$ of Section 2.2 is written here for a general step size $h > 0$, and $\Omega_{j,h} = \{x \in \Omega : x + he_j \in \Omega \text{ and } x - he_j \in \Omega\}$ denotes the points whose shifted evaluations remain in the domain. A *trimming weight* is a measurable function $w : \Omega \to [0, 1]$ with $\operatorname{supp}(w) \subseteq \Omega_{j,h}$; it confines the finite difference to points where both shifted arguments lie inside the support, so that the prediction function is never evaluated where the regression function is unidentified. The following conditions are maintained throughout.

Assumption 1. The data are i.i.d. draws from P with $\mathbb{E}[Y^2] < \infty$ and $\operatorname{Var}(Y \mid X = x) \le \bar{\sigma}^2 < \infty$ for P-almost every x. The covariate distribution admits a density p on Ω with $0 < p_{\min} \le p(x) \le p_{\max} < \infty$ for all $x \in \Omega$.

The density bounds are standard in the average derivative literature (Powell et al., 1989; Newey and Stoker, 1993) and enter the theory only through the ratio $p_{\max}/p_{\min}$,

which bounds the Riesz representer above. They are genuinely restrictive, and one design of interest violates them: independent Gaussian covariates have a density that is not bounded below on $\mathbb{R}^d$, and the representer of Lemma 1 is then unbounded. That case is covered by an alternative to the density bounds, under which every result of this section continues to hold with the sup-norm conditions replaced by moment conditions.

Assumption 2 (Unbounded support)**.** In place of the density bounds of Assumption 1: (a) $p(x) > 0$ for every $x \in \Omega$; (b) there exists $r > 1$ with

$$\mathbb{E}\left[\left(\frac{p(X - he_j)}{p(X)}\right)^r\right] < \infty \quad \text{and} \quad \mathbb{E}\left[\left(\frac{p(X + he_j)}{p(X)}\right)^r\right] < \infty \tag{3.1}$$

and (c) $\alpha_h \in L^q(P)$ for some $q > 2$. The moment conditions on Y in Assumption 1 are retained unchanged.

Condition (b) is precisely what the density bounds were doing in the proofs. Every appeal to $p_{\min}$ below reduces to controlling a shifted second moment, $\mathbb{E}[g(X \pm he_j)^2] = \mathbb{E}[g(X)^2\, p(X \mp he_j)/p(X)]$, which a uniform density bound handles by brute force and which (3.1) handles by Hölder, at the cost of a slightly higher integrability requirement on g. Condition (c) replaces the sup-norm bound that Lemma 1 supplies in the bounded case; it is an assumption here rather than a conclusion, because with an unbounded density the explicit formula for α_h need not be bounded even though it remains square integrable.

Section 5 exhibits both assumption sets: its Gaussian designs satisfy Assumption 2 while failing Assumption 1, and a bounded-support arm with covariates drawn uniformly on a box satisfies Assumption 1 literally and carries a representer bounded by $1/(2h)$. Coverage agrees between them to within four thousandths, so nothing in what follows depends on which route is taken. The verification for the Gaussian case — the shift ratio $\exp(hu_j - h^2/2)$, which has moments of every order, and the representer $\alpha_h(u) = e^{-h^2/2}\sinh(hu_j)/h$, which is square integrable but unbounded — is given in Section S12.

3.3 The Window Estimand and Its Riesz Representation

Definition 1 (Window Average Marginal Effect)**.** Fix $h > 0$ and a trimming weight w with $\text{supp}(w) \subseteq \Omega_{j,h}$. The *window average marginal effect* of variable j at scale h is:

$$\theta_h(f) = \mathbb{E}\left[w(X)\,\Delta_{j,h} f(X)\right] \tag{3.2}$$

Up to the trimming weight, $\theta_h(f)$ is exactly the population counterpart of the quantity the estimator of Section 2 computes: the finite difference at step h is not a numerical approximation to a derivative that the theory then ignores, but the defining operation of the estimand itself. The step size is a fixed feature of the estimand, chosen and reported by the analyst exactly as a bandwidth is, and the default $h_j = 0.05\,\hat{\sigma}_j$ of Section 2 corresponds to asking how the predicted outcome responds to displacements of a twentieth of a standard deviation — an economically interpretable perturbation scale. Two features deserve emphasis. First, $\theta_h(f)$ is well defined for any bounded measurable f: no differentiability is assumed anywhere, which is what permits a single estimand to cover neural networks and regression trees simultaneously. Second, the trimming weight repairs a defect of the untrimmed implementation, which evaluates $\hat{f}$ at points $x_i \pm h e_j$ that may fall outside the support of the covariates, where the regression function is not identified; w excises a boundary shell of probability mass $O(h)$ and confines every evaluation to the region where f means something.

The weighting in Definition 1 is unconditional, and it is worth being explicit about what that means, because the trimming is not always the negligible boundary adjustment it is for most features. The definition integrates $\Delta_{j,h} f$ against w without renormalizing, so $\theta_h(f)$ is the *weighted* average effect over the whole population and not the average effect among the retained observations. When w is an indicator the two are related by

$$\theta_h(f) = \mathbb{E}[w(X)] \cdot \mathbb{E}\left[\Delta_{j,h} f(X) \,\middle|\, w(X) = 1\right], \tag{3.3}$$

so the conditional effect among retained observations is $\theta_h(f)/\mathbb{E}[w(X)]$, and the two coincide only when $w \equiv 1$. The unconditional form is the one the theory treats, and

deliberately so: it is linear in f, which is what delivers the exact Riesz representation of Section 3.3 and the exact product bias of Section 3.4. The ratio is not linear, and an estimator for it would carry a second influence-function term for the denominator.

The estimand admits an exact dual representation, which is the engine of everything that follows. Extend $q = wp$ by zero outside Ω.

Lemma 1 (Riesz Representation)**.** *Let Assumption 1 hold and define:*

$$\alpha_h(u) = \frac{q(u - he_j) - q(u + he_j)}{2h\, p(u)}, \qquad u \in \Omega \tag{3.4}$$

Then for every $g \in L^1(P)$:

$$\mathbb{E}\left[w(X)\, \Delta_{j,h} g(X)\right] = \mathbb{E}\left[\alpha_h(X)\, g(X)\right] \tag{3.5}$$

Moreover $\mathbb{E}[\alpha_h(X)] = 0$, and under the density bounds of Assumption 1, $\|\alpha_h\|_\infty \leq p_{\max}/(h\, p_{\min})$. Under Assumption 2 the representer need not be bounded, and (3.5) holds for every g with $\alpha_h g \in L^1(P)$.

The proof is a change of variables in each half of the finite difference and nothing more. Two features of (3.5) carry the rest of the paper. No boundary term arises, because the trimming weight keeps every shifted evaluation point inside Ω, so the density need not be assumed to vanish on the boundary as it must be for the derivative functional; and the representer involves the density only through ratios of its values at points $2h$ apart, so no density derivative is ever estimated. The mean-zero property follows by applying (3.5) to $g \equiv 1$. The cost of the exactness is the factor h^{-1} in the bound: smaller windows mean noisier representers. Section S1 compares the construction with the integration-by-parts representation of the classical average derivative, which is where the boundary term and the density score come from.

The classical estimand is recovered in the small-window limit.

Proposition 1 (Recovery of the Average Derivative)**.** *Let $f \in W^{1,1}(\Omega)$ and let w be a*

trimming weight with $\operatorname{supp}(w) \subseteq \Omega_{j,h_0}$ *for some* $h_0 > 0$. *Then:*

$$\lim_{h \downarrow 0} \theta_h(f) = \mathbb{E}\left[w(X)\, \partial_j f(X)\right] \tag{3.6}$$

If moreover $(w_m)_{m \geq 1}$ *is a sequence of trimming weights with* $w_m \uparrow 1$ P*-almost everywhere, then* $\mathbb{E}[w_m\, \partial_j f] \to \mathbb{E}[\partial_j f] = \mathrm{AME}_j$.

One weak derivative — the minimal regularity under which the classical estimand exists at all — is thus the only condition separating the window effect from the average derivative. Nothing about the fitted model appears in Proposition 1: the double limit concerns the estimand, not the estimator, and in practice one fixes h and reports the window effect, treating the proposition as the licence to interpret it as a derivative summary when the truth is smooth.

3.4 Orthogonal Score, Cross-Fitting, and Efficient Inference

The Riesz representation converts the estimand into a moment condition with an exact second-order bias. Define, for candidate functions $\tilde{f}, \tilde{\alpha}$ and scalar θ, the score:

$$\psi(x, y; \tilde{f}, \tilde{\alpha}, \theta) = w(x)\, \Delta_{j,h} \tilde{f}(x) + \tilde{\alpha}(x)\left(y - \tilde{f}(x)\right) - \theta \tag{3.7}$$

At the truth, $\mathbb{E}[\psi(X, Y; f, \alpha_h, \theta_h(f))] = 0$: the first term has mean $\theta_h(f)$ by definition and the correction term has conditional mean zero. The content of the construction is what happens away from the truth.

Lemma 2 (Exact Product Bias)**.** *Let Assumption 1 hold and let* $\tilde{f} \in L^2(\Omega)$, $\tilde{\alpha} \in L^2(P)$. *Then:*

$$\mathbb{E}\left[\psi(X, Y; \tilde{f}, \tilde{\alpha}, \theta_h(f))\right] = \mathbb{E}\left[\left(\alpha_h(X) - \tilde{\alpha}(X)\right)\left(\tilde{f}(X) - f(X)\right)\right] \tag{3.8}$$

In particular the bias is bounded by $\|\alpha_h - \tilde{\alpha}\|_{P,2} \cdot \left\|\tilde{f} - f\right\|_{P,2}$.

Lemma 2 is an identity, not an expansion: because θ_h is linear in f, the moment bias equals the product of the two nuisance errors with no linearization remainder whatsoever.

Neyman orthogonality — the vanishing of the Gateaux derivative of the moment in each nuisance direction at the truth (Chernozhukov et al., 2018) — is the special case obtained by setting one error to zero, but the exact form is stronger and is what makes the subsequent analysis short. The practical meaning: an error in the fitted learner harms the estimate only to the extent that the representer is also misestimated, and vice versa. It is this multiplication of errors, not any smoothness of the learner, that buys the parametric rate.

The estimator combines the score with cross-fitting. Partition $\{1, \ldots, n\}$ into K folds $I_1, \ldots, I_K$ of approximately equal size ($K = 5$ is the package default), and for each k let $\hat{f}^{(-k)}$ and $\hat{\alpha}^{(-k)}$ be estimates of f and α_h computed from the observations outside fold k — the learner exactly as in Section 2, the representer as described in Section S5. The debiased estimator is:

$$\hat{\theta}_h = \frac{1}{n} \sum_{k=1}^{K} \sum_{i \in I_k} \left[w(\mathbf{x}_i)\, \Delta_{j,h} \hat{f}^{(-k)}(\mathbf{x}_i) + \hat{\alpha}^{(-k)}(\mathbf{x}_i) \left(y_i - \hat{f}^{(-k)}(\mathbf{x}_i) \right) \right] \tag{3.9}$$

Cross-fitting decouples the randomness of the nuisance estimates from the observations at which the score is evaluated, so that no entropy or Donsker condition on the learner's function class is needed (Chernozhukov et al., 2018) — a point of substance here, since the function classes of boosted trees and deep networks are exactly the classes for which such conditions fail.

Assumption 3. For each fold k: (a) $\left\| \hat{f}^{(-k)} - f \right\|_{P,2} = o_p(1)$ and $\left\| \hat{\alpha}^{(-k)} - \alpha_h \right\|_{P,2} = o_p(1)$; (b) $\left\| \hat{f}^{(-k)} - f \right\|_{P,2} \cdot \left\| \hat{\alpha}^{(-k)} - \alpha_h \right\|_{P,2} = o_p(n^{-1/2})$; (c) $\left\| \hat{\alpha}^{(-k)} \right\|_{\infty} \leq \bar{A} < \infty$ almost surely.

Condition (c) is without loss of generality: α_h obeys the known bound of Lemma 1, so any estimate may be truncated at that bound without degrading its L^2 error. Conditions (a) and (b) are rate conditions on *regression-type* L^2 errors only. The sufficient condition for (b) that we verify for specific learners in Section S5 is that both nuisances converge faster than $n^{-1/4}$; the product form also permits trading accuracy in one nuisance against the other.

Under Assumption 2 in place of the density bounds, (a) strengthens to an $L^{2+\delta}$ re-

quirement and (c) to a qth moment bound, while the product condition (b) — the one that governs how fast the nuisances must converge — is unchanged. That it is unchanged is a consequence of the exactness of Lemma 2: the bias is a product of two L^2 errors with no linearization remainder, so the step of the argument that carries the rates never touches a density bound, and the rate conditions of Section S5 apply verbatim under either assumption set. Theorems 1 and 2 and Proposition 2 hold in both cases, and the proofs carry the two side by side at the three points where they differ. Section S1 states the modified conditions and shows that the strengthening of (a) is free whenever the learner's error is uniformly bounded.

Theorem 1 (Asymptotic Linearity and Normality)**.** *Let Assumptions 1 and 3 hold, write* $\psi_i = \psi(\mathbf{x}_i, y_i; f, \alpha_h, \theta_h(f))$*, and suppose* $V_h = \mathrm{Var}(\psi_i) > 0$*. Then:*

$$\sqrt{n}\left(\hat{\theta}_h - \theta_h(f)\right) = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \psi_i + o_p(1) \xrightarrow{d} N(0, V_h) \tag{3.10}$$

Moreover the plug-in variance estimator:

$$\hat{V}_h = \frac{1}{n} \sum_{k=1}^{K} \sum_{i \in I_k} \left[w(\mathbf{x}_i)\, \Delta_{j,h} \hat{f}^{(-k)}(\mathbf{x}_i) + \hat{\alpha}^{(-k)}(\mathbf{x}_i) \left(y_i - \hat{f}^{(-k)}(\mathbf{x}_i)\right) - \hat{\theta}_h \right]^2 \tag{3.11}$$

satisfies $\hat{V}_h \xrightarrow{p} V_h$*.*

No smoothness of the fitted learner appears anywhere in Theorem 1 or its proof. The only property of the finite difference operator that is used is that it is bounded on $L^2(P)$ after trimming:

$$\|w\, \Delta_{j,h} g\|_{P,2} \leq \frac{1}{h} \left(\frac{p_{\max}}{p_{\min}}\right)^{1/2} \|g\|_{P,2} \tag{3.12}$$

so L^2 convergence of the learner transfers to L^2 convergence of its window effects automatically — precisely the step that fails for the derivative operator, which is unbounded on every L^p space, and the reason the classical route requires function-space convergence of the learner's gradient. The condition that the earlier version of this theory imposed at this point — root-n convergence of $\hat{f}_n$ to f in a Sobolev or BV norm — is not merely weakened but absent: it has been replaced by Assumption 3(b), a product of L^2 rates

that growing-complexity learners can attain.

The exactness of Lemma 2 yields a second dividend beyond weakened rates: the score is doubly robust, and one direction of that robustness removes the consistency requirement on the learner altogether. Setting $\tilde{\alpha} = \alpha_h$ in Lemma 2 makes the moment bias vanish identically for *every* $\tilde{f}$ — not approximately, not at first order, but as an identity — and cross-fitting converts this into exact finite-sample unbiasedness of $\hat{\theta}_h$. The learner's failures are then paid entirely in variance, never in location.

Proposition 2 (Double Robustness: Inference without Learner Consistency)**.** *Let Assumption 1 hold and suppose that for each fold: (a)* $\left\|\hat{f}^{(-k)} - g\right\|_{P,2} = o_p(1)$ *for some fixed* $g \in L^2(\Omega)$*, not necessarily equal to* f*; (b)* $\left\|\hat{\alpha}^{(-k)} - \alpha_h\right\|_{P,2} = o_p(n^{-1/2})$ *— in particular* $\hat{\alpha}^{(-k)} = \alpha_h$ *when the covariate density is known; and (c)* $\left\|\hat{\alpha}^{(-k)}\right\|_\infty \leq \bar{A} < \infty$ *almost surely. Define* $m_g = w\,\Delta_{j,h} g + \alpha_h\,(f - g)$*, which satisfies* $\mathbb{E}[m_g(X)] = \theta_h(f)$ *for every* g*, and:*

$$V_h(g) = \mathrm{Var}\big(m_g(X)\big) + \mathbb{E}\big[\alpha_h(X)^2\,\sigma^2(X)\big] \tag{3.13}$$

If $V_h(g) > 0$*, then:*

$$\sqrt{n}\,\big(\hat{\theta}_h - \theta_h(f)\big) \xrightarrow{d} N\big(0, V_h(g)\big), \qquad \hat{V}_h \xrightarrow{p} V_h(g) \tag{3.14}$$

so the confidence intervals of Corollary 1 retain asymptotic coverage. When $\hat{\alpha}^{(-k)} = \alpha_h$ *exactly,* $\mathbb{E}[\hat{\theta}_h] = \theta_h(f)$ *at every sample size.*

The reach of Proposition 2 is wider than its hypotheses may suggest, because condition (a) is what off-the-shelf defaults actually satisfy. Fixed-complexity learners — a depth-capped forest, an early-stopped boosting run, a network of fixed width — converge not to f but to a pseudo-true limit g in their approximation class, so Assumption 3(a) fails and Theorem 1 is silent about them. The proposition covers exactly that case whenever the representer side is strong: in simulation designs and designed experiments α_h is known in closed form, and in the binary case of Remark S2 a randomized X_j means a known propensity score, so the marginal effect of a randomized treatment is validly covered for any black-box learner with no consistency requirement at all — the model-agnostic

analogue of the classical robustness of randomization (Robins et al., 1994).

Two qualifications. When $g \neq f$ the estimator is asymptotically linear with an influence function other than the efficient one; there is no conflict with Theorem 2, because such an estimator is not regular, and efficiency is restored exactly when $g = f$. And in observational data the density is unknown, so condition (b) demands a parametric-rate representer, which Riesz regression delivers only under strong structure. The proposition is a complement to Theorem 1 rather than a replacement: the theorem splits the burden multiplicatively across the two nuisances, the proposition shifts it entirely onto one. Section S1 records the mirror-image direction, under which a parametric-rate learner tolerates an arbitrarily poor representer estimate.

Theorem 2 (Semiparametric Efficiency)**.** *Let Assumption 1 hold. In the nonparametric model in which the distribution of (X, Y) is restricted only by Assumption 1, the efficient influence function of the functional $P \mapsto \theta_h(f_P)$ is $\psi(x, y; f, \alpha_h, \theta_h(f))$, and the semiparametric efficiency bound is:*

$$V_h = \mathrm{Var}\big(w(X)\,\Delta_{j,h} f(X)\big) + \mathbb{E}\big[\alpha_h(X)^2\,\sigma^2(X)\big] \tag{3.15}$$

where $\sigma^2(x) = \mathrm{Var}(Y \mid X = x)$. The estimator $\hat{\theta}_h$ of Theorem 1 attains the bound.

The variance decomposition (3.15) is the honest accounting of what inference on a marginal effect costs. The first term is the heterogeneity of the effect itself across the covariate distribution — irreducible, present even if f were known. The second is the price of not knowing f: the conditional noise amplified by the squared representer, which scales as h^{-2} in the worst case through the bound of Lemma 1. Narrow windows purchase proximity to the derivative estimand at the cost of a larger efficiency bound — the bias–variance tradeoff of the problem, located in the estimand rather than hidden in an estimator's tuning. No estimator that is regular for θ_h can do better than (3.15); the question the earlier version of this theory could not answer — whether the procedure is optimal in any sense — has the sharpest available affirmative answer for the window estimand.

Corollary 1 (Confidence Intervals)**.** *Under the conditions of Theorem 1,* $\hat{\theta}_h \pm z_{1-\gamma/2}\,(\hat{V}_h/n)^{1/2}$ *is an asymptotically valid* $1-\gamma$ *confidence interval for* $\theta_h(f)$*, and the associated* t*-statistic is asymptotically standard normal under* $H_0 : \theta_h(f) = 0$*.*

3.5 What Learner Gradients Estimate, and Which Learners Qualify

Two questions remain, separate from one another and from the inference theory above: what the uncorrected plug-in average of a learner's own finite differences actually estimates, and what must be checked before applying the results above to a learner not treated here. Both are developed in full in Sections S3–S6 of the supplement; this subsection records what they establish, because the answers bear on how the empirical results below should be read.

For estimators of bounded variation — which includes every tree-based method, however its splits are chosen — a pairing identity characterizes the window effect exactly. Write $q = wp$ for the trimmed covariate density and let $K_h \star q$ denote its one-dimensional moving average along coordinate j, the smoothing the window itself induces:

$$(K_h \star q)(u) = \frac{1}{2h}\int_{-h}^{h} q(u + s\,e_j)\,ds. \tag{3.16}$$

Theorem 3 (Pairing Identity)**.** *Let Assumption 1 hold, let* w *be a trimming weight with* $\mathrm{supp}(w) \subseteq \Omega_{j,h}$*, and let* $g \in BV(\Omega)$ *(identified with its precise representative). Then:*

$$\mathbb{E}\left[w(X)\,\Delta_{j,h}g(X)\right] = \int_{\Omega} (K_h \star q)(u)\; dD_j g(u) \tag{3.17}$$

where $D_j g$ *is the* j*-th component of the distributional gradient of* g*. In particular, for a piecewise constant* g *with leaf partition* $\{R_l\}$ *and leaf values* $\{\bar{y}_l\}$*, by Lemma S3:*

$$\mathbb{E}\left[w(X)\,\Delta_{j,h}g(X)\right] = \sum_{l\sim m} (\bar{y}_m - \bar{y}_l)\,(\nu_{lm})_j \int_{\partial R_l \cap \partial R_m} (K_h \star q)\; d\mathcal{H}^{d-1} \tag{3.18}$$

A piecewise constant fit has no pointwise derivative anywhere, so (3.18) is the sense

— and so far as we are aware the first explicit one — in which a tree ensemble's marginal effect is a well-defined population quantity rather than an artefact of the differencing step: the total jump of the fitted surface across leaf boundaries in direction j, each boundary weighted by its share of the covariate mass in an h-window. In the small-window limit it reduces to the usual average derivative whenever the fitted surface has one (Corollary S1).

Whether the plug-in is *consistent* for that quantity is a separate, function-space question. Neural networks with smooth activations converge in $W^{1,p}$, where Rellich–Kondrachov compactness upgrades L^p consistency of the fit to convergence of its gradient (Theorem S1); Mondrian forests and L_2Boost with a fixed symmetric linear smoother live in BV, where the upgrade comes from BV compactness (Theorems S3 and S2). Breiman forests and gradient boosting with greedily grown trees — both used in Section 4 — fall outside these results, the first conjectured, the second not covered at all. Neither is a gap in the inference theory: Theorem 1 consumes an $L^2(P)$ rate and nothing else, and Theorem 3 already characterizes what the window effect of any bounded-variation learner measures. What is unsettled is only whether their *uncorrected* plug-in is consistent for the gradient, and the evidence of Section 5 is that for the boosted ensemble it is not — precisely the case the correction exists to handle. Table S1 of the supplement records the position for these classes and for several that are not, together with the rate conditions and the two obligations a new learner must meet.

4 Empirical Examples

Of the three applications introduced in Section 1, the credit default data are reported in full here, being the sharpest of the three for the purpose at hand. Adult Income and Ames Housing are reported in Sections S10 and S11 of the supplement; the design, the model classes and the window settings set out below are common to all three, and the conclusions drawn from the credit default data hold in the other two.

All three applications fit the same four model classes — logistic or linear regression as the parametric baseline, a random forest, XGBoost, and a feedforward neural network —

and estimate the window effect by the debiased cross-fitted procedure of Section 2.4 with $K = 5$ folds: within each fold the learner and the Riesz representer are fit on the other four and evaluated on the held-out fold, so no learner is ever differenced at an observation it was trained on, and standard errors come from the sample variance of the orthogonal score. Architectures, hyperparameters, training schedules and the handling of predicted probabilities are common to the three applications and are recorded in Section S7 of the supplement.

The window estimand requires two settings to be fixed before estimation, and because $\theta_{j,h}$ is defined by them they are reported rather than treated as tuning. The step size is the default $h_j = \max(10^{-4}, 0.05\hat{\sigma}_j)$, floored at 0.5 for integer-valued features. The floor matters for count variables such as bedrooms, bathrooms, years of education and months delayed: a count has no mass strictly between its levels, so a window narrower than one unit has an empty interior, and for a piecewise constant learner the difference across it vanishes except where the window happens to straddle a split. At $h_j = 0.5$ the window is exactly the one-unit contrast, which is also the quantity of substantive interest for a count. The second setting is the trimming weight, which gives weight zero to observations lying within h_j of the boundary of the observed support of x_j, where $x \pm h_j e_j$ would require the learner to extrapolate. For most features this removes a negligible share of the sample, but for variables with mass sitting exactly on a boundary it does not: capital gains and capital losses are zero for over ninety percent of the Adult sample, and zero is the minimum of their observed support, so those features are retained for 7.7% and 4.7% of observations respectively. By (3.3) the reported effects are the effects among that small minority scaled by its population share, not the effects on the full sample and not the conditional effects among the retained; dividing by the retained share recovers the latter. They should not be read against the untrimmed features in the same column without that adjustment. Table S2 of the supplement reports h_j and the trimmed fraction for every feature in all three applications, and the discussion below flags the cases where the trimming is large enough to change the interpretation.

4.1 UCI Credit Default

The second dataset is the UCI Default of Credit Card Clients dataset (Yeh, 2009; Dua and Graff, 2019), comprising 30,000 observations of Taiwanese credit card holders. The outcome is whether a cardholder defaulted on their payment in the following month. Features are organized into demographics (age, sex, education, marital status), payment history (average payment status, months delayed), and bill and payment amounts (average bill amount, average payment amount, payment ratio). The specification search considers logistic regression and a TensorFlow neural network, providing a direct comparison between a classical parametric model and a deep learning approach on the same inferential task. Two features here carry substantial trimming. Months delayed takes seven integer values and 70.9% of the sample sits on one of the two outer ones, so its effect is identified among cardholders strictly inside that range; education takes four levels and is trimmed at 36.8%. Both take the floored step size $h_j = 0.5$, so their reported effects are one-level contrasts. The payment ratio, which turns out to carry the largest effect in the application, is continuous and trimmed at 18.4%.

A specification search over the three covariate blocks — demographics, payment history, and billing and payment amounts — is reported in Section S8 of the supplement. Its conclusions are that payment history dominates the demographic block, that the payment ratio is much the largest effect in the design once it is available, and that the ordering and approximate magnitude of the effects are stable across specifications, while the demographic effects shrink substantially once payment history is controlled for. The remainder of this section works with the full specification.

Table 1 compares all four models on the full specification, and it is the sharpest illustration in the paper of what the orthogonal correction changes. Under the plug-in the four models flatly contradicted one another on the two variables that matter. Months delayed was 0.083 for logistic regression, 0.018 for the random forest, 0.021 for XGBoost and 0.112 for the neural network, a sixfold spread; the payment ratio was -0.431 and -0.440 for the two tree ensembles, -0.045 for logistic regression, and 0.0009 — indistinguishable from no effect at all — for the neural network. A researcher reading

that table had to conclude either that the models were detecting genuinely different structure, or that at least three of the four were wrong, with no way to tell which.

The debiased estimates resolve the contradiction rather than splitting the difference. Months delayed becomes 0.030, 0.013, 0.019 and 0.025, and the payment ratio becomes -0.265, -0.489, -0.584 and -0.194: every model now agrees that a higher payment ratio is associated with a substantially lower probability of default, including the neural network, whose plug-in estimate had put the effect at zero. The demographic panel converges the same way. The female AME was -0.020, -0.006, -0.015 and -0.014 under the plug-in and is -0.0198, -0.0161, -0.0175 and -0.0176 under the correction; married was 0.022, 0.005, 0.014 and 0.019 and is now 0.0206, 0.0222, 0.0205 and 0.0186. The variation that the earlier draft read as four models weighing the covariates differently was in the main the regularization bias $\mathbb{E}[\alpha_h(\hat{f} - f)]$, which differs across learners because their regularization differs, and which the correction removes. What survives is a common estimate of a common target.

Convergence is not complete, and the payment ratio shows where it stops. The four debiased estimates still span -0.194 to -0.584, a range far wider than their standard errors of roughly 0.07, so the models have not been brought into agreement to within sampling error on that variable. Payment ratio is the most skewed feature in the design and is trimmed at 18.4%, and the second-order product term that Theorem 1 requires to be $o_p(n^{-1/2})$ is plausibly not negligible here for the weaker nuisance fits. The correct reading is that the correction removes the first-order bias and leaves a residual that the theory bounds but does not eliminate, which is visible precisely where the nuisance problem is hardest. Finally, the random forest standard errors, which the plug-in bootstrap reported as 0.0000 to four decimal places on the female indicator, are 0.0045 under the influence-function formula — essentially identical to the other three models, as they should be for a common estimand at a common sample size.

The same comparison against SHAP values and partial dependence slopes is reported for this application in Section S9 of the supplement, and for the other two applications alongside it. Its findings follow the pattern described in Section 1: the SHAP column

is on a different scale from the effect column and in several places carries the opposite sign, because it answers a question about attribution rather than about a rate of change; and the partial dependence column, being a plug-in average over a single full-sample fit, agrees with the debiased effect where the underlying plug-in is nearly unbiased and separates from it towards zero wherever regularization flattens the fitted surface — on the payment ratio, the hardest nuisance problem in the design, by a factor of six or more for every learner.

5 Simulation

We evaluate the estimator through three simulation studies, which together separate what the window average recovers from what can be said about it. Simulation 1 examines the bias and root mean squared error of the uncorrected plug-in window average as a function of sample size, across three data generating processes and two outcome types. Simulation 2 asks whether the refitting bootstrap of Section 3.4 delivers valid intervals for that plug-in, comparing empirical coverage of nominal 95% intervals against the target. Simulation 3 repeats both exercises for the debiased cross-fitted estimator of Section 3.4, holding the designs, learners and windows fixed so that the two estimators differ only in the orthogonal correction. The three are best read in order: the first locates a bias, the second shows that resampling cannot see it, and the third removes it. All features x_1, x_2, x_3, x_4 are drawn independently from $N(0,1)$. Features x_1 and x_2 have nonzero true effects; x_3 and x_4 are pure noise with true AME of zero. Three data generating processes are considered. The linear DGP sets $\eta = 2x_1 + 3x_2$. The nonlinear DGP sets $\eta = 2x_1^2 + 3x_2$, so that the true AME of x_1 is zero in the regression case by symmetry of $N(0,1)$ — $\mathbb{E}[4x_1] = 0$ — but nonzero in the classification case where the sigmoid breaks that symmetry. The interaction DGP sets $\eta = 2x_1 + 3x_2 + 2x_1x_2$, which preserves the marginal AMEs of x_1 and x_2 at 2.0 and 3.0 respectively in the regression case since $\mathbb{E}[2x_2] = \mathbb{E}[2x_1] = 0$. For regression, the outcome is $y = \eta + \varepsilon$ where $\varepsilon \sim N(0,1)$. For classification, $P(y = 1) = \sigma(\eta)$ where σ denotes the logistic function. True AMEs for all DGPs and outcome types are

computed via Monte Carlo integration using n = 1,000,000 observations drawn from the true prediction function directly, treating the result as ground truth. Sample sizes of $n \in \{250, 500, 1\,000, 2\,500, 5\,000\}$ are evaluated over 500 Monte Carlo iterations for classification and 1,000 iterations for regression.

The simulation study fits four model classes with hyperparameters chosen to balance computational tractability against representativeness of typical applied use. The random forest uses 100 trees with a maximum depth of 6, and XGBoost uses 100 boosting rounds with a maximum depth of 4 and a learning rate of 0.05 — conservative settings that prevent overfitting on the small sample sizes considered while remaining flexible enough to capture the nonlinear and interaction structures in the data generating processes. The neural network is a feedforward architecture with two hidden layers of 32 and 16 units, trained for 50 epochs with batch size 64 and learning rate 10^{-3} using the Adam optimizer. For regression outcomes, the network includes the input normalization layer and target standardization wrapper described in Section 2 to ensure scale-invariant gradient updates. Logistic and linear regression are included as baselines and use no hyperparameter tuning. All model fits are performed independently on each Monte Carlo iteration with no warm starting between iterations, so the variance estimates reflect the full sampling distribution of the AME estimator including model fitting variability. The Monte Carlo ground truth AMEs are computed once at the start of the study using n = 1,000,000 observations drawn from the true prediction function directly, with the resulting AMEs treated as known quantities for the bias and coverage calculations. Simulation 2 uses 1,000 bootstrap resamples per Monte Carlo iteration and 1,000 Monte Carlo iterations per cell for the linear, logistic, random forest and XGBoost learners. The neural network uses 200 resamples and 500 iterations: every resample is a fresh training run, which makes the network roughly nine tenths of the total cost of the study, and the calibration result the argument rests on is the forest's. Percentile intervals are used throughout, which is why the resample count is set well above the couple of hundred that would suffice for a standard error alone. Simulation 3 uses 1,000 Monte Carlo iterations per cell for every learner; it needs no resampling, since its standard errors come from the influence function.

Table 2 collects the three studies on the linear data generating process, the one design on which all three report results, so that the plug-in, the refitting bootstrap and the debiased estimator can be compared like for like on a common target. The per-feature detail, the intermediate sample sizes, the nonlinear and interaction designs, and the sieve-representer and bounded-support arms are reported in Section S12 of the supplement, together with the discussion of each individual table. The pattern in those designs is the one the summary shows.

Simulation 1 asks whether the uncorrected plug-in average recovers the window effect, and the answer separates the learners by how heavily they are regularized rather than by how flexible they are. Linear and logistic regression, XGBoost and the neural network all converge on the truth: their absolute bias falls to 0.001 or below on the continuous outcome and to 0.0005 or below on the binary one by $n = 5{,}000$. XGBoost is the worst of them at $n = 250$, at 0.362 on the continuous outcome, which is the familiar small-sample underfitting of a boosted ensemble and is a rate problem that disappears rather than a limit problem that does not. The random forest is the exception, and it is the instructive one. Its bias does not shrink but grows, from 0.059 at $n = 250$ to 0.093 at $n = 5{,}000$ on the continuous outcome, and holds near 0.016 throughout on the binary one. This is regularization bias in the sense of Section 3.4: the depth and subsampling constraints impose a shrinkage that does not vanish with the sample, so the plug-in converges to something other than $\theta_{j,h}$ and no amount of data repairs it.

Simulation 2 asks whether the refitting bootstrap delivers valid inference for that plug-in, and it does not. The failure takes two forms, and the coverage column records both. Where the plug-in carries the non-vanishing bias just described, the interval is centred in the wrong place and coverage collapses as the sample grows and the interval narrows around the wrong point: the forest covers at 0.092 on the continuous outcome and 0.186 on the binary one at $n = 5{,}000$, having covered at 0.870 and 0.968 at $n = 250$. Growing the sample makes this failure worse, not better. The second form is milder and affects even a correctly specified model: because each replicate is evaluated at the observed sample rather than at the resample, the bootstrap distribution omits the sampling variability of

the average of the derivative over the covariates, so the standard errors are too small — logistic regression covers at 0.781 on its largest effect — and the network over-corrects to 1.000 on the continuous outcome, intervals so wide they are uninformative. Neither failure is visible in the bootstrap output itself, which is the practical point: both produce intervals that look unremarkable.

Simulation 3 repeats the design for the debiased cross-fitted estimator of Section 3.4, holding the data generating processes, learners, hyperparameters, sample sizes, window and Monte Carlo ground truth at the values used in the two preceding studies, so that any difference is attributable to the orthogonal correction and to nothing else. The covariates are independent standard normals, so the representer of Lemma 1 is available in closed form, $\alpha_h(u) = e^{-h^2/2}\sinh(hu_j)/h$, and no representer estimation is required; a second arm replaces it with a sieve estimate, and a third uses bounded support, where the trimming weight is active rather than vacuous. The correction removes both failures at once. Absolute bias is at most 0.003 at $n = 250$ and at most 0.0005 at $n = 5{,}000$ for every learner on both outcome types, and for the forest it falls from 0.093 to 0.000 at the largest sample — the correction does not merely shrink the bias at a given sample size but restores the $O(n^{-1/2})$ rate, which is what inference requires. Coverage lies between 0.938 and 0.972 across every learner and both outcome types, against a bootstrap range of 0.092 to 1.000 on the same designs and the same ground truth. The learners that were already nearly unbiased are essentially unchanged, as they should be: the orthogonal score corrects a first-order term that is near zero for them.

The closed-form representer is a luxury of the design, so the arm that estimates it is the one that bears on practice. Replacing the closed form with a sieve estimate and holding everything else fixed, coverage survives with a mild conservatism — mean 0.958 over the same cells, against the closed form's 0.948 — and the cost appears where the theory says it should, in the product bias of Assumption 3(b). Mean absolute bias is 3.3 times the closed form's at $n = 250$, then 1.6, 1.2 and 1.1, and 0.9 at $n = 5{,}000$: indistinguishable by the largest sample. Nothing in the procedure requires the representer to be known, and the price of not knowing it is a finite-sample one that the rate conditions of Section 3.4

already anticipate.

6 Conclusion

This paper has proposed a framework for interpreting any supervised machine learning model through average marginal effects, and the central claim is that the traditional accuracy–interpretability tradeoff is not fundamental. The move that carries the argument is to make the window the estimand rather than a numerical approximation to something else: the window average marginal effect is the average centred difference of the regression function at a fixed, analyst-chosen step, it is well defined for a learner that is differentiable nowhere, and it recovers the classical average derivative in the small-window limit whenever one weak derivative exists. A single estimand therefore covers trees, ensembles and networks without differentiability assumptions of any kind, and the estimator returns a coefficient table — estimate, standard error and significance level — directly analogous to parametric regression output, from the estimated influence function in a single pass. Existing interpretability methods, SHAP (Lundberg and Lee, 2017), LIME (Ribeiro et al., 2016) and partial dependence plots (Friedman, 2001) among them, do not provide that.

Because the estimand is linear in the regression function it admits an exact Riesz representation by a change of variables, with no boundary terms and no density derivatives, and the orthogonal score built from that representer has estimation bias exactly equal to a product of two L^2 nuisance errors. The cross-fitted estimator is therefore root-n consistent, asymptotically normal under the $n^{-1/4}$-type rates that flexible learners can satisfy, and efficient; a double robustness result extends validity to inconsistent, fixed-complexity learners whenever the representer is known or parametrically estimable. On the interpretation side a pairing identity characterizes the population quantity that a tree-based model's marginal effect measures, and gradient consistency holds for neural networks in $W^{1,p}$ and for Mondrian forests in the space of functions of bounded variation, delineating the regime in which the uncorrected plug-in is already consistent. That regime is

narrower than it may appear, and its boundary is instructive rather than merely technical: at its own risk-minimizing number of rounds a greedily boosted ensemble can carry substantially more total variation than the function it is estimating, so its uncorrected plug-in gradient is in part measuring its own roughness. That is the regularization bias of Section 3.4 seen from the interpretation side, and an argument for the correction rather than a gap in it.

Empirically the four model classes agree far more closely under the debiased estimator than under the plug-in, and they do so because they are now estimating the same quantity rather than each reporting the slope of its own fit. The credit default application shows it most sharply: plug-in estimates of the payment-ratio effect ranging from -0.431 to 0.0009 across the four learners — a set that cannot all be describing the same world — become four significant negative estimates telling one story. Standard errors move in the direction the theory predicts, which in several places means upward, the refitting bootstrap having reported 0.0000 to four decimal places for the random forest where the influence-function standard error for the same quantity is orders of magnitude larger: narrow intervals around a displaced centre, the failure mode isolated in Simulation 2. The cost of the correction is visible in the smallest application, where with $n = 1{,}460$ observations several Ames Housing effects that the plug-in reported as precisely determined are no longer distinguishable from zero, and the estimator reports that rather than concealing it.

The most pressing open problem concerns Breiman random forests (Breiman, 2001b), which produce the least stable plug-in estimates of any class considered and which the framework splits into two questions of very different difficulty. The inferential question needs only an L^2 rate, since Theorem 1 consumes nothing else; consistency is available under structural assumptions (Scornet et al., 2015), and converting it into a rate meeting the $n^{-1/4}$ benchmark — or exploiting the product form of Assumption 3(b) with a fast representer estimate — is a concrete and plausibly tractable target. Simulation 3 indicates how much is already settled in the case the theory handles cleanly: with the representer known, Proposition 2 asks nothing of the forest beyond convergence to some fixed limit, and a depth-constrained forest whose plug-in error grows with the sample

size nevertheless yields intervals whose coverage averages 0.948. The interpretive question remains genuinely hard, since data-dependent splits couple partition geometry to response variation in a way that prevents the key decomposition. Separating the two is itself progress: the most widely used forest implementation can be brought under the inferential guarantees without waiting for the harder interpretation theory. Beyond it, the estimand as defined here is predictive rather than causal, and the finite-sample behaviour of the procedure depends on choices the asymptotic theory does not pin down — the number of folds, the truncation level for the estimated representer, the trimming weight near the boundary, and above all the window h, which is part of the estimand and whose sensitivity should be reported the way bandwidth sensitivity is.

Data availability

All three datasets analysed in this paper are publicly available. The Adult Income and the Default of Credit Card Clients datasets are distributed by the UCI Machine Learning Repository (Dua and Graff, 2019), and the Ames Housing dataset is distributed with De Cock (2011). Code reproducing every numerical result in the paper and in the supplement — the three simulation studies, the three empirical applications, and the scripts that generate each table — is available at `https://github.com/penny4nonsense/marginfx-paper`, together with the download scripts for the raw data and the environment specification needed to run them. The estimator itself is implemented in the `marginfx` package. A permanently archived version of the code, with a digital object identifier, will be deposited before publication.

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Table 1: UCI Credit Card Default: Model Comparison. Average marginal effects on P(Default), full specification (A+B+C).

Feature	Logistic	Random Forest	XGBoost	Neural Net
age	0.0001	-0.0002	-0.0005	-0.0000
	(0.0003)	(0.0003)	(0.0003)	(0.0003)
female	-0.0198***	-0.0161***	-0.0175***	-0.0176***
	(0.0046)	(0.0045)	(0.0045)	(0.0045)
education	-0.0040*	-0.0018	0.0034	-0.0040*
	(0.0022)	(0.0022)	(0.0022)	(0.0022)
married	0.0206***	0.0222***	0.0205***	0.0186***
	(0.0063)	(0.0062)	(0.0062)	(0.0063)
avg_pay_status	-0.0218***	0.0061	0.0907***	0.0040
	(0.0075)	(0.0074)	(0.0075)	(0.0072)
months_delayed	0.0301***	0.0134***	0.0191***	0.0254***
	(0.0012)	(0.0011)	(0.0011)	(0.0012)
avg_bill_amt	-0.0000**	-0.0000***	-0.0000***	-0.0000*
	(0.0000)	(0.0000)	(0.0000)	(0.0000)
avg_pay_amt	0.0000	-0.0000***	-0.0000***	-0.0000***
	(0.0000)	(0.0000)	(0.0000)	(0.0000)
pay_ratio	-0.2651***	-0.4890***	-0.5836***	-0.1943**
	(0.0730)	(0.0670)	(0.0685)	(0.0773)
McFadden R^2	0.137	0.210	0.194	0.166
Accuracy	0.803	0.821	0.815	0.808

Notes: ***p<0.01, **p<0.05, *p<0.10. Estimates are the debiased cross-fitted window AME with $K = 5$ folds; standard errors from the orthogonal score in parentheses. Step sizes h_j and trimmed fractions are reported in Table S2.

Table 2: Summary of the three simulation studies on the linear data generating process. Simulation 1 reports the bias of the uncorrected plug-in average, Simulation 3 the bias of the debiased cross-fitted estimator, and the final two columns the empirical coverage of nominal 95% intervals at $n = 5{,}000$ under the refitting bootstrap of Simulation 2 and under the influence function of Simulation 3.

	Plug-in |bias|		Debiased |bias|		Coverage at $n = 5{,}000$	
Learner	$n = 250$	$n = 5{,}000$	$n = 250$	$n = 5{,}000$	Bootstrap	Influence fn.
Panel A: continuous outcome, true AMEs 2.000 and 3.000						
Linear	0.003	0.000	0.003	0.000	0.956	0.960
Random forest	0.059	0.093	0.002	0.000	0.092	0.946
XGBoost	0.362	0.053	0.002	0.001	0.503	0.938
Neural net	0.095	0.001	0.003	0.000	1.000	0.959
Panel B: binary outcome, true AMEs 0.199 and 0.298						
Logistic	0.0083	0.0005	0.0011	0.0002	0.781	0.940
Random forest	0.0163	0.0158	0.0018	0.0003	0.186	0.944
XGBoost	0.0616	0.0047	0.0027	0.0003	0.877	0.972
Neural net	0.0120	0.0002	0.0011	0.0002	0.894	0.942

Notes: Bias entries are the mean of |bias| over the two signal features x_1 and x_2, in the units of the outcome; the noise features x_3 and x_4 have a true effect of zero and are within Monte Carlo error of it under every learner and both estimators. Coverage entries are whichever signal feature's coverage lies furthest from the nominal 0.95, so that over-coverage is reported as well as under-coverage. The plug-in and bootstrap columns are the uncorrected estimator of Simulations 1 and 2; the debiased and influence-function columns are the cross-fitted estimator of Simulation 3 with $K = 5$ folds and the closed-form representer. Full per-feature results are in the supplement.